

DANCING WITH THE STARS: MEASURING THE ROTATION CURVE OF THE MILKY WAY GALAXY

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Abstract

We obtained radio observations of the 21-cm emission line from neutral hydrogen (HI) to test the existence of dark matter in our own galaxy. Using the UW Small Radio Telescope (SRT), we obtained HI spectra at Galactic longitudes 22° – 72° and applied a geometric method to extract the orbital velocity of HI clouds and their corresponding distance from the Galactic center. We then constructed a rotation curve and the enclosed mass profile, and compared results with predictions from luminous matter alone. Our rotation curve shows a plateau in velocity, though classical physics expects the rotation curve to decline with radius due to the distribution of visible matter in the galaxy. These results suggest the existence of dark matter and confirm trends found in previous published studies.

Introduction

Dark matter remains one of the great mysteries in astrophysics. The unexpected shape of galactic rotation curves created by Vera Rubin provided the first insights into the presence of dark matter. In this work, we construct galactic rotation and mass interior curves and evaluate their agreement with published results. Figure 1 displays the Galactic rotation curve published in Bhattacharjee et al. (2014).

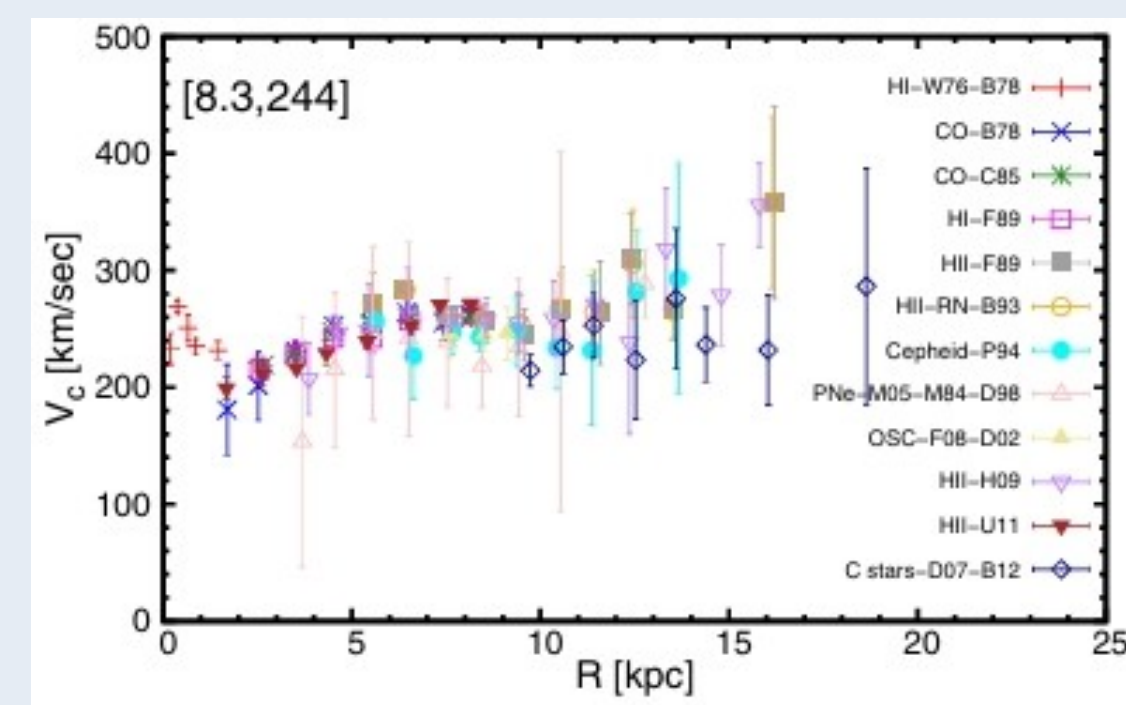


Fig. 1: Rotation Curve of the Milky Way out to 200kpc

Observations

All data was collected using the Sterling Hall Small Radio Telescope during November 2025. We observed the Galactic HI 21 cm line using the built-in spectrometer. A bandpass filter centered at 1420.4 MHz, corresponding to the emission line of neutral hydrogen, was used in mode 4 (156 bins of frequency). Eleven locations within the Galactic Plane were observed, with Galactic longitude selected every 5° and ranging from $[20^\circ, 72^\circ]$. Group 1 started at 22° and integrated for 400 seconds, and Group 2 started at 20° and used 500 seconds.

Figure 2 shows a visualization of the Tangent-point Method. The Tangent-point Method is used to choose points where calculation of the Galactocentric radius is simplified due to the geometry. Tangent points are located in the Galactic plane, on the circle that passes through both the Galactic Center and the Sun.

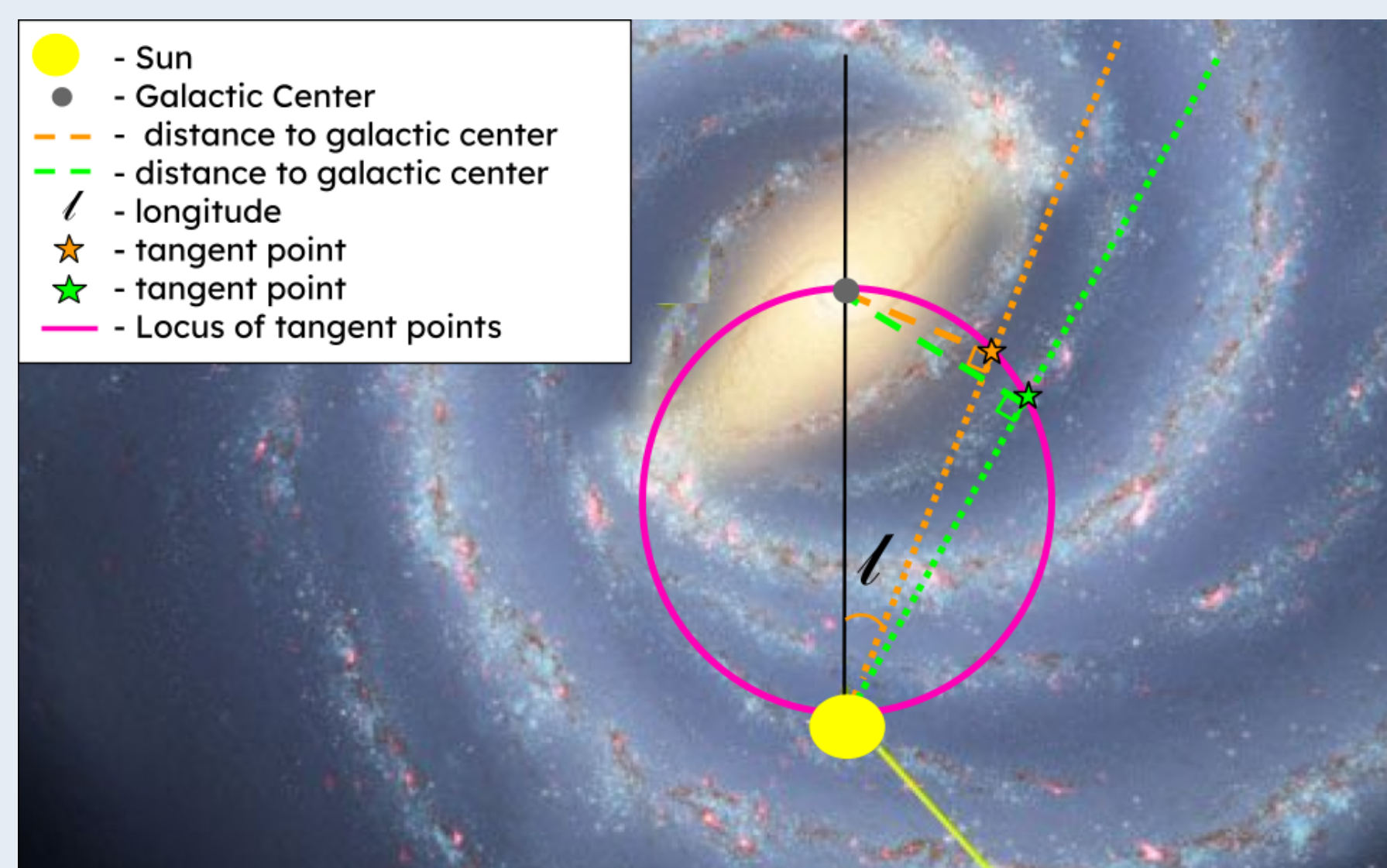


Fig. 2: Tangent Point Method. Photo credit NASA JPL

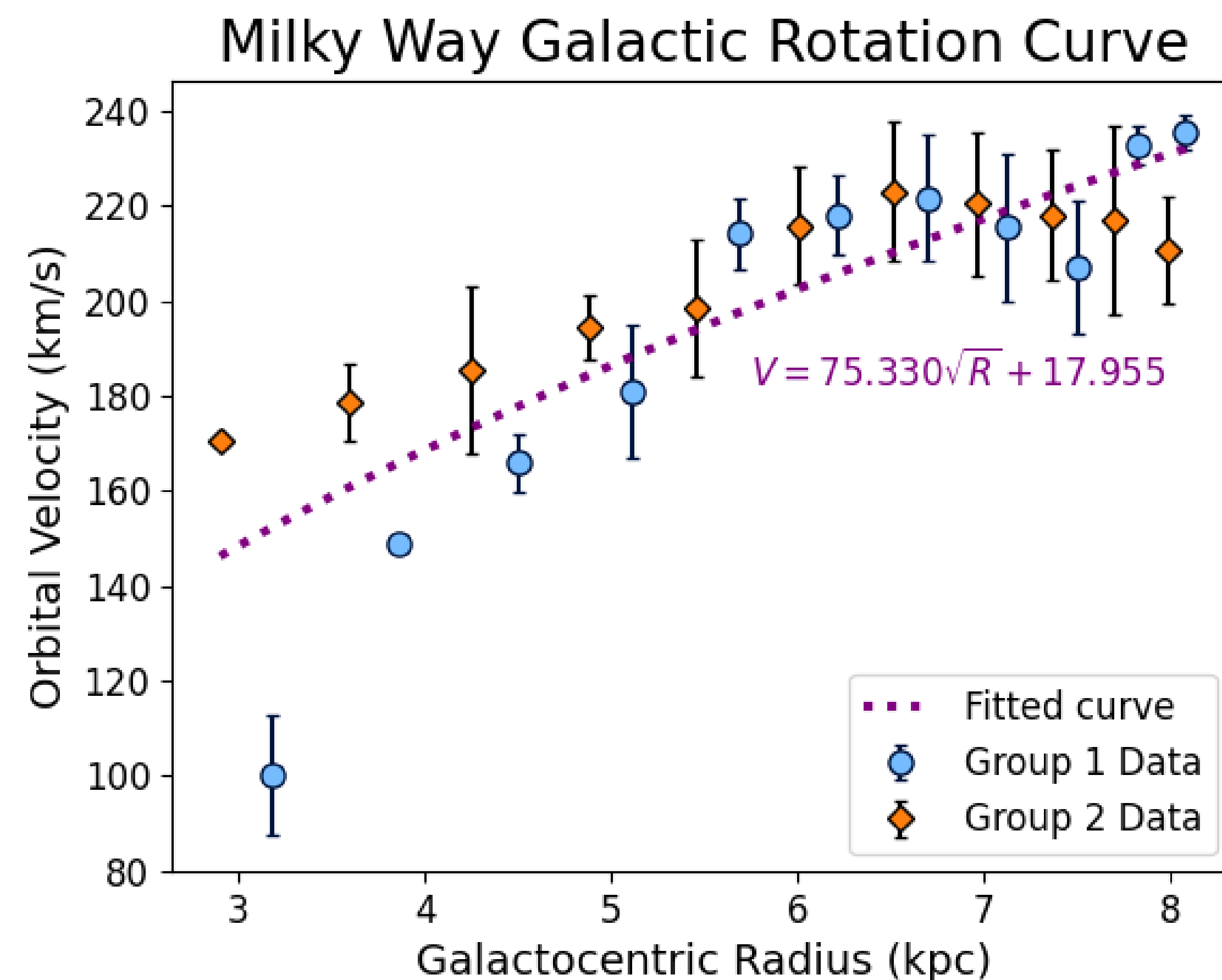


Fig. 3: Galactic rotation curve of the Milky Way Galaxy

Data & Methods

Figure 4 presents an example spectrum at a Galactic longitude of 52° , before and after data processing. We averaged the spectra at each position, converted frequency to topocentric radial velocity with the non-relativistic Doppler shift equation, and fit a linear baseline using sections of the plot without signal. The third panel of Figure 4 shows the resulting spectrum after subtracting the baseline out and converting to a standard LSR Velocity Frame.

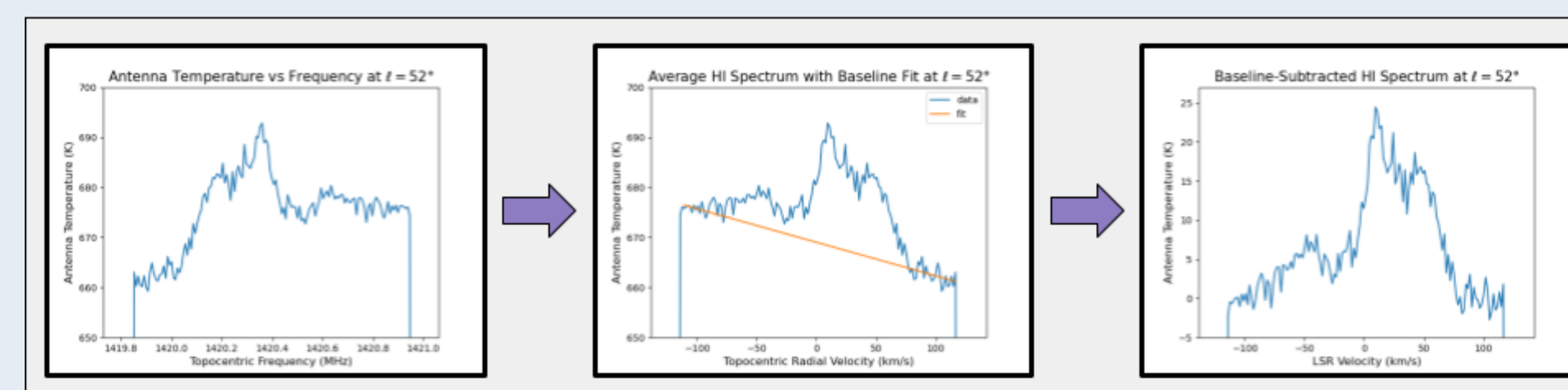


Fig. 4: Example spectrum before and after data processing steps.

For each tangent point, the equations for the orbital radius R and the velocity V of an object about the Galactic Center are as follows:

$$R = R_\odot \sin l \quad (1)$$

$$V = V_r + V_\odot \sin l \quad (2)$$

where $R_\odot$ is the Galactocentric radius of the Sun, $V_\odot$ is the orbital velocity of the Sun about the Galactic Center, l is the longitude, and V_r is the highest radial velocity where signal is present, which we calculated for each individual longitude.

References

- Bhattacharjee et al. (2014). *Rotation Curve of the Milky Way Out to 200 kpc*. *The Astrophysical Journal*, p.63.
- Gustaf et al. (2025). *Astronomy 465 UW SRT Laboratory 5 data*.
- Marr et al. (2015). *Fundamentals of Radio Astronomy: Observational Methods*.
- R. Hurt, NASA, JPL *Artistic Impression of the Milky Way*
- NEROC Haystack Observatory Undergraduate Research Initiative: *Small Radio Telescope*. MIT.
- UW-Madison SRT Manual*. Compiled by S. Stanimirović.

Results

We plot orbital velocity as a function of Galactocentric radius in order to create the Milky Way Galactic rotation curve. Data from two groups observing at different Galactic longitudes is plotted in Figure 3.

From these orbital velocity and Galactocentric radius values we can calculate the mass interior to each radius,

$$M_R = \frac{V^2 R}{G}. \quad (3)$$

Enclosed mass is plotted as a function of radius in Figure 5.

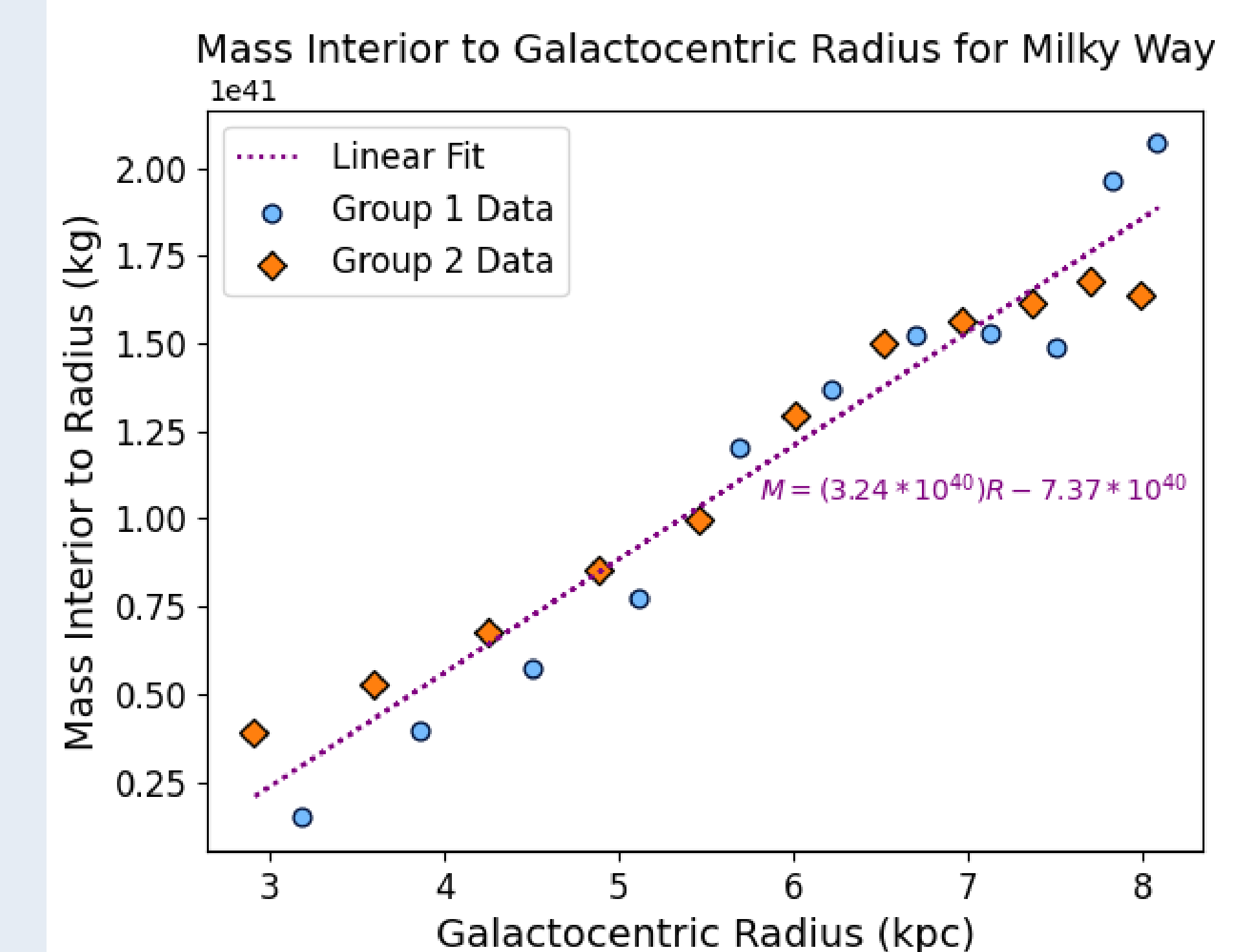


Fig. 5: Enclosed mass profile of Milky Way Galaxy.

Conclusion

Our measured rotation curve matches the curve published in Bhattacharjee et al. (2014) (Figure 1), contradicting what is expected from luminous matter alone. Classically, one would expect a gentle decline in galactic rotation velocity after about 4 kpc, approaching 150 km/s; however, our measurements show an approximate plateau even after this point. This discrepancy is precisely what implies the existence of dark matter. Additionally, our estimate of the mass contained within the Milky Way does not match the expected mass distribution predicted by visible matter, further supporting the existence of dark matter.

Inevitably, we understand that there are errors present in our measurements arising from systematic or analytical errors, including thermal noise, but the estimated uncertainties are not significant enough to alter our results. In future work, we could observe for a longer time at each position to reduce the thermal noise and obtain more data points.

Dark matter is an area of active research. Discovering its properties could allow us to understand the behavior of our Galaxy and many others.

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